



Zamson Lim Zhen Pin

FINAL-YEAR CYBERSECURITY STUDENT · SOC & DETECTION ENGINEERING

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Final-year BSc (Hons) Cybersecurity student at Asia Pacific University. I joined an employer that had been hit by ransomware months earlier with no monitoring in place, proposed a Security Operations Centre, got approval, and **built it as the only person on security** in four months. **Malware development for purple-team testing is the part I care most about** — writing the payload and detonating it in an isolated lab tells you exactly what your detections see and what they miss, and it is where most of my rules came from.

Looking for: a graduate SOC analyst or detection engineering role.

EXPERIENCE

Network & Security Administrator (Intern)

May 2026 – Aug 2026

Palmgold Management Sdn Bhd · Kuala Lumpur · sole owner of security operations

BUILT

- **The company's Security Operations Centre, in four months**, from a proof of concept on a spare desktop to the stack that now runs the company's monitoring — the answer to a ransomware incident the previous August that had left no logs to investigate.
- **Deployed a Wazuh SIEM with co-located OpenSearch** on a Proxmox host I set up myself, integrating firewall, core and access switching, wireless, Windows endpoint, Active Directory and Zeek telemetry through custom ingest pipelines.
- **Rolled out monitoring agents to 80+ Windows machines** via Active Directory Group Policy, using an idempotent installer that was safe to re-run and logged per machine.
- **Built the network sensor** — switch port mirroring into a multi-node Zeek cluster through a Python decapsulation stage — giving visibility into internal traffic the firewall never sees.

DETECTION & PURPLE TEAM

- **Wrote the detection content:** custom decoders and correlation rules matched to the hardware actually in use, deliberately without an endpoint-agent dependency I had no authority to deploy fleet-wide.
- **Ran authorized purple-team exercises**, attacking from Kali Purple then detecting from the SIEM, and turned each into detection content; surfaced an unmonitored network segment that no rule could have covered.
- **Ran an authorized internal vulnerability assessment** and delivered three audience-tailored reports, honestly downgrading one finding to "unconfirmed" when it could not be reproduced.

OPERATIONS

- **Recovered a full ingestion outage** in which archive logging filled the disk and silently flipped every index read-only, then installed retention policies so it could not recur.
- **Handed over a documented, running SOC** at the end of the internship.

SKILLS

SIEM	Wazuh (deployment, agent management, custom decoders and correlation rules), OpenSearch query DSL and aggregations, ingest pipeline design, file integrity monitoring, alert triage and incident response
NETWORK	Zeek network security monitoring, FortiGate policy and syslog integration, Aruba switching and wireless, traffic mirroring and decapsulation, Wireshark, Cisco Packet Tracer
SYSTEMS	Proxmox virtualization, Ubuntu / Rocky Linux administration, systemd tuning, Nginx reverse proxy and TLS, Windows Server, Active Directory and Group Policy at fleet scale
SECURITY	Purple-team exercise design and execution, authorized vulnerability assessment (OpenVAS, Nmap, Nikto), malicious code analysis in isolated labs, Autopsy forensics, Kali Linux
CODE	Python, JavaScript (D3.js, Three.js), Bash, Windows batch, Java, C++, C#, Rust
LANGUAGES	English, Mandarin, Malay, Cantonese

BSc (Hons) Cybersecurity

2024 – 2027

Asia Pacific University · Kuala Lumpur · final year

- Coursework across network security, digital forensics and secure development.

PROJECTS**DETECTION & PURPLE TEAM****Purple team, both chairs**

- Ran the red side from Kali Purple — a C2 listener behind a tunnelling service, implants beaconing on a fixed interval — then switched sides and hunted it from the SIEM.
- Building the samples is the part I enjoy most: writing the payload and detonating it in an isolated lab tells you exactly what your detections see and what they miss.
- The firewall and the tunnel showed only a shared address; DNS was the only thing that named the infected machines. A plain implant to an ordinary server was **missed**.

Ransomware canary detection

- File integrity monitoring rules scoped to a canary folder, mapped to MITRE T1486, to catch mass-encryption behaviour early.
- Validated with a benign encryptor hard-guarded to that folder — never a live sample.
- Path-scoping was the fix after fleet-wide scoping fired on every agent's overnight scan and spammed the whole estate.

COLLECTION & INFRASTRUCTURE**Zeek network monitoring pipeline**

- Mirrored traffic decapsulated in Python and replayed onto a virtual interface into a multi-node Zeek cluster.
- Detections for domain-generation bursts, DNS tunnelling, exfiltration and inbound scans.
- Re-characterised an alarming reported capture loss as an MTU artefact of the mirror tunnel — which decides which detections can be trusted.

TOOLING & FORENSICS**SOC operator dashboards**

- Three single-file consoles built for daily use: a D3 world traffic map, an endpoint alert console, and a Three.js traffic visualizer.
- De-identified and published as open-source templates; live demos and source linked from the portfolio.

Capture The Flag

2024 – present

SekaiCTF · CYNX · BeSec · picoCTF

- SekaiCTF 2025 with team Digital Phantoms: solved a medium reverse-engineering challenge on an Android APK using jadx-gui, keytool and Python.

Active Directory monitoring

- Learned AD properly — domain structure, Group Policy, event log semantics — then built monitoring on native Windows event logs.
- Saved searches for authentication failure, log clearing and account enumeration, ranked by volume so an unusual spike surfaces early.

Authorized vulnerability assessment

- Internal assessment from Kali Purple using OpenVAS, with Nmap, Nikto and command-line work to verify by hand what the scanner claimed.
- Findings included unauthenticated printer administration, exposed print services and a reverse-proxy bypass on an office device.
- Delivered as three reports for three audiences; re-scanned and downgraded one finding to “unconfirmed” when it would not reproduce.

Alerting pipeline

- Custom Python integration pushing critical alerts to chat, so triage never depended on somebody watching a dashboard.
- De-duplicated with file locking and checked against the API response, and tolerant of binary content that would otherwise break the send.

Wireless & access point monitoring

- Wireless syslog brought into the SIEM with custom decoders for association, authentication and roaming events.
- A scheduled enrichment pipeline resolving hardware addresses to hostnames, so an alert names a machine rather than a MAC.

Malicious file analysis (Autopsy)

- Forensics exercise at APU examining a lecturer-provided sample in Autopsy.
- Worked through the file's contents and metadata rather than trusting a scanner verdict; written up as a formal report.